

1. Equilibrium / 1 bath / finite temp. magnetism.

→ model: $n := n_a + n_b$, $m := n_a - n_b$; $n=1$ fixed, $t_a = -t_b$

$$\begin{pmatrix} \epsilon_a(k) + \frac{V}{2}n - \frac{V}{2}m & -\Delta^* \\ -\Delta & \epsilon_b(k) + \frac{V}{2}n + \frac{V}{2}m \end{pmatrix}$$

cond. shift.

bare bands: $\epsilon_b(k) = \pm \frac{G}{2} - 2t_a(\cos(k_x) + \cos(k_y))$

$G \neq$ fixed gap! → see: <https://www.desmos.com/calculator/azrtesau5i>

bare band gap: $\epsilon_a(0) - \epsilon_b(0) = G - 2t_a(1+1) + 2t_b(1+1) = G - 4(t_a - t_b)$

↳ $t_a = -t_b \Rightarrow G_0 = G - 8t_a$

Hasse bands (= diry brs Δ) U_n v kodu

$$\epsilon_a^H - \frac{V}{2} = \epsilon_a(k) \pm \frac{V}{2}m$$

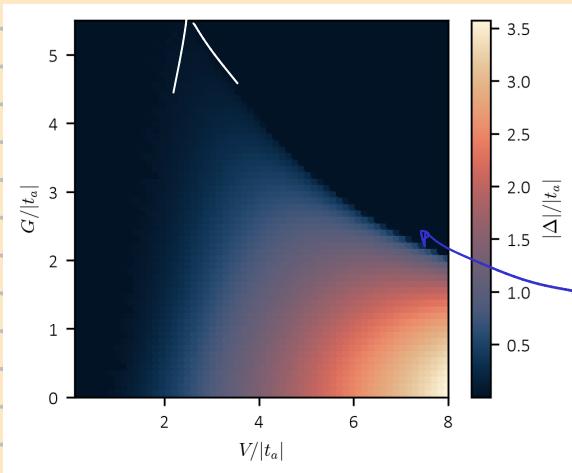
Hasse gap: $\epsilon_a^H(0) - \epsilon_b^H(0) = G - 8t_a - V \cdot m \equiv G_n$

note: $t_a \neq t_b$ is gap opens for $E_c = \frac{V}{2}n + \frac{t_a + t_b}{t_a - t_b} \cdot \frac{Vm - G}{2}$
 $\uparrow$ $V = U_n$ divided by m !

19. In model $U_n = V$ $\begin{pmatrix} \epsilon_a + U_n n_b & \dots \\ \dots & \epsilon_b + U_n n_a \end{pmatrix} \rightarrow \Delta = 0$ vedno za $\nexists$ EC režim!

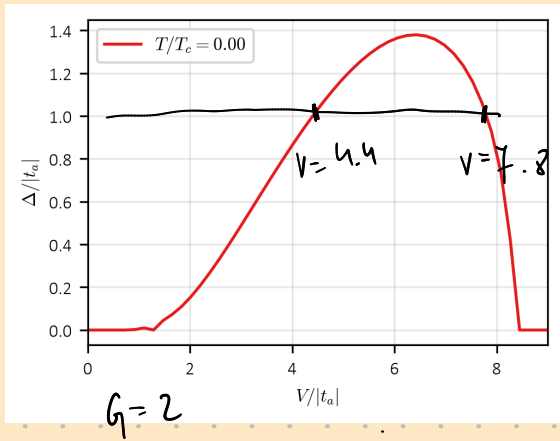
↳ če $U_n < V$ ($U_n = 0 =$ BCS teorija) $\Rightarrow \Delta > 0$ tudi za $G_0 > 0$.

$G_{max} \sim 5$
 Z_n
 $U_n = V$



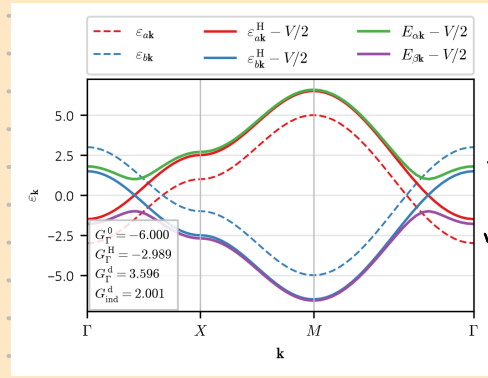
Hasse shift
 vedno pozitivna
 za Δ , in
 ne ravni.
 Manjša $\neq$ več V , ker
 Hasse potegne navzgor.

prime 1: $V_H = V$:

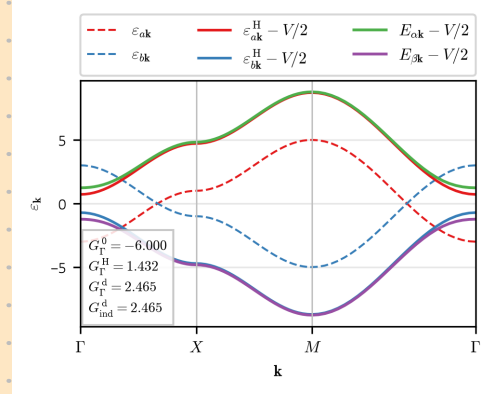


$V=4.4$

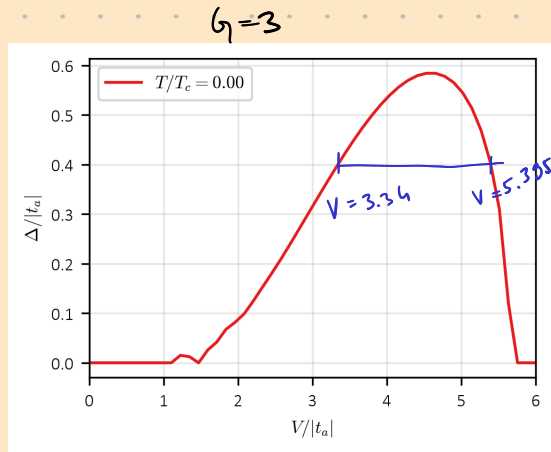
$V=7.8$



$\downarrow -\frac{V}{2}$
 $\delta_0 = G_{\Gamma}^0$

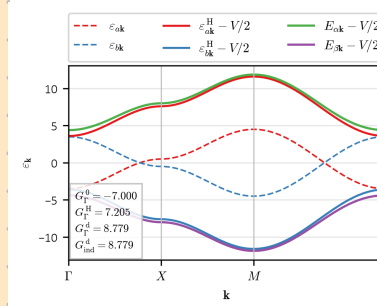
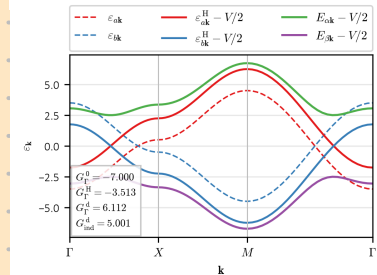
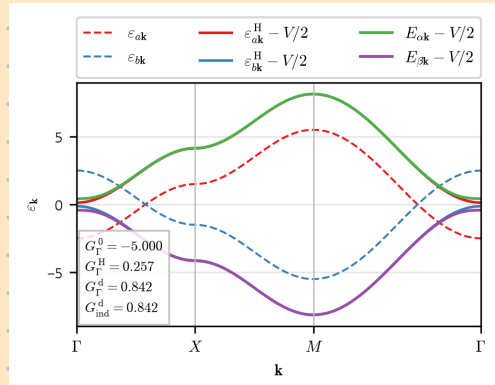
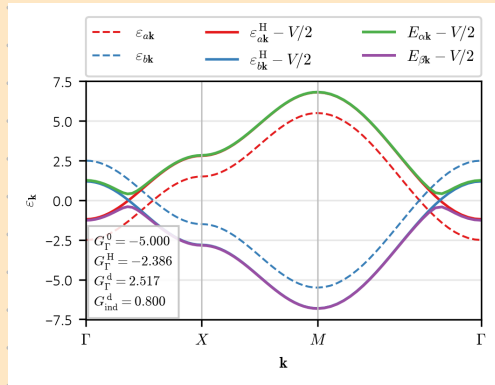


BEC-2m (Hawke!)

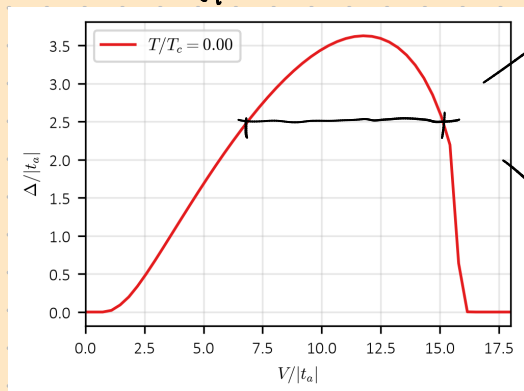


$V=3.34$

$V=5.4$



$G=10$

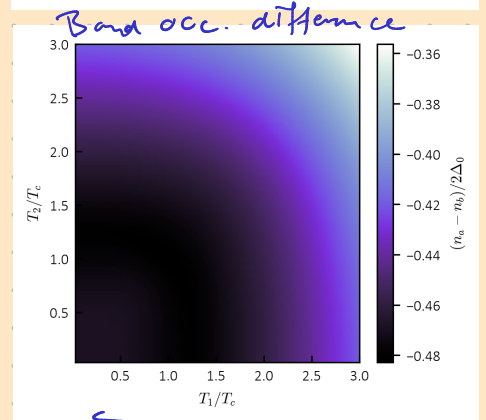
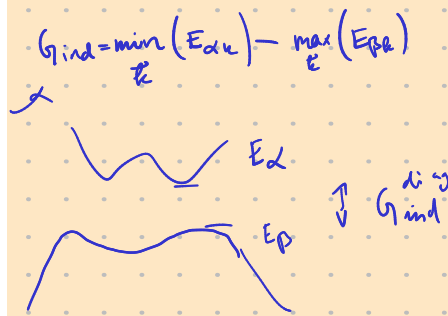
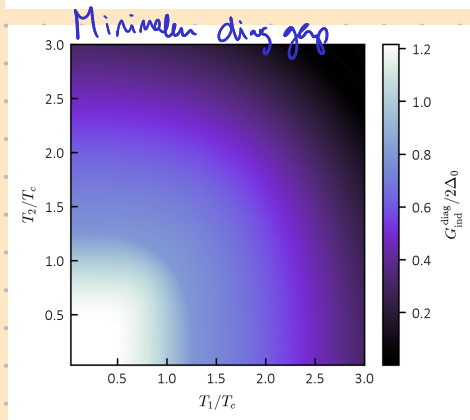
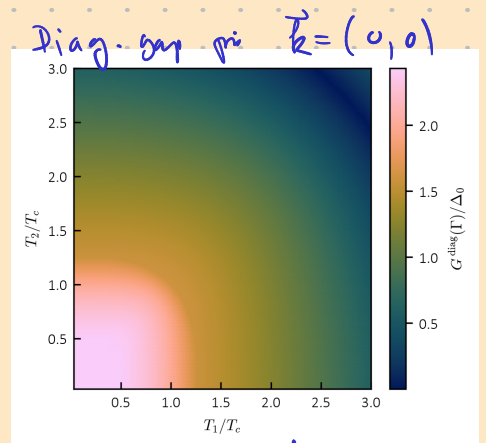
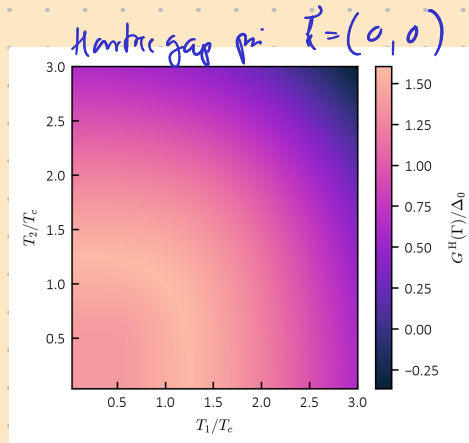
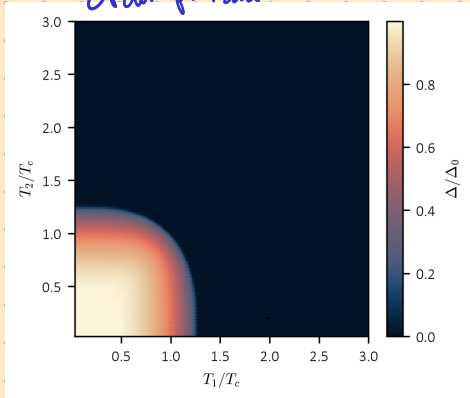


$V=6.866$

$V=15.13$

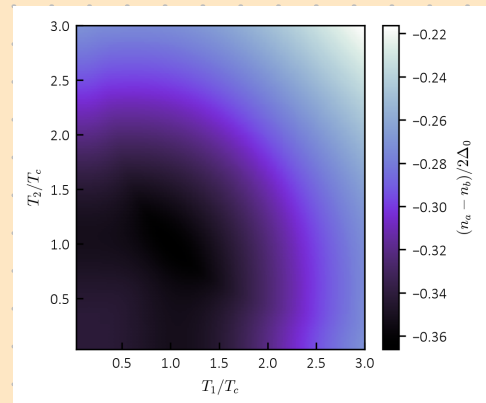
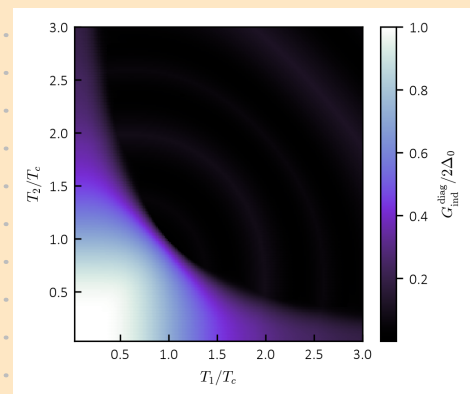
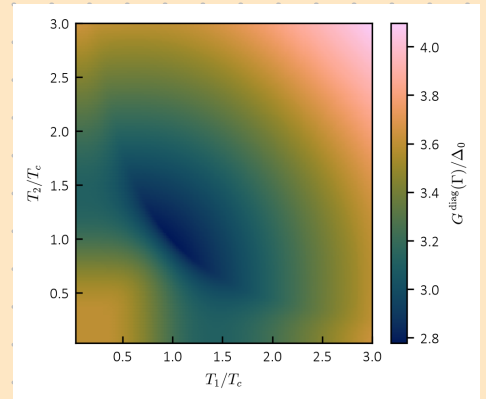
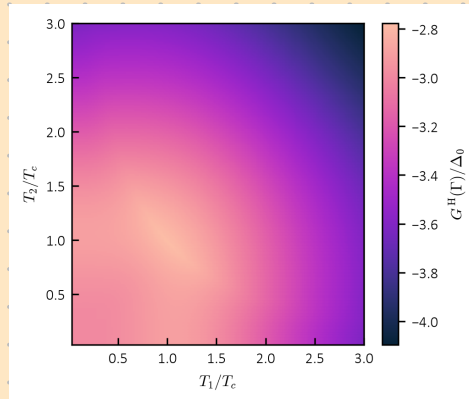
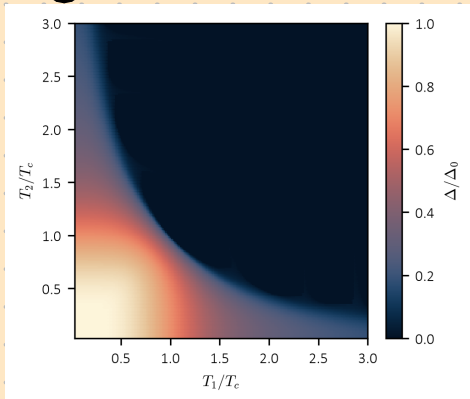
② GAPS maps zu: $G=2$, kopul: $\gamma(T) = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \cdot \begin{matrix} \text{xx} \\ \text{pp} \end{matrix} \cdot \text{r.r.} \cdot f\left(\frac{E_{\text{xx}} - \frac{\gamma}{2}}{T}\right) f\left(\frac{-E_{\text{pp}} - \frac{\gamma}{2}}{T}\right)$

Ⓐ $V=7.8$ (BEC limit):
order param.

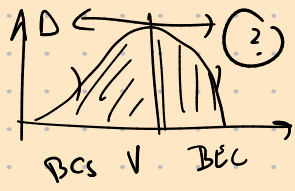


bei niedriger Temperatur
bei hoher Temperatur

Ⓑ $V=4.4$ (BCS limit bei idealem Δ)



3.) kaj se dogaja vmes?



okoli $\Delta \approx 0$:

$$\Delta' = \Delta \cdot \Lambda$$

$$\Lambda = V \sum_k m_k \mu_k v_k (n_{\beta k} - n_{\alpha k}) = \frac{V}{N} \sum_k \frac{n_{\beta k} - n_{\alpha k}}{E_{\alpha k} - E_{\beta k}} (= |\xi_k| \cdot 2)$$

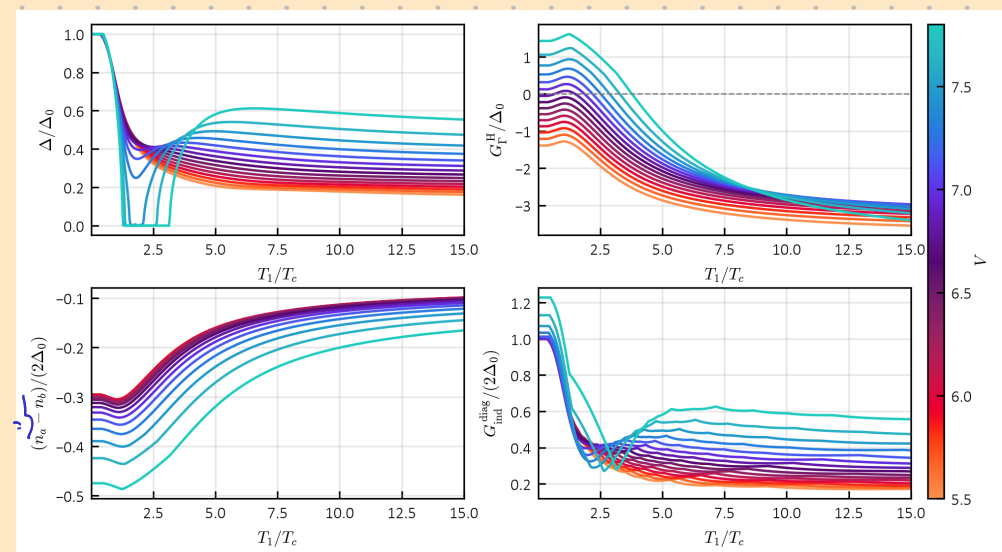
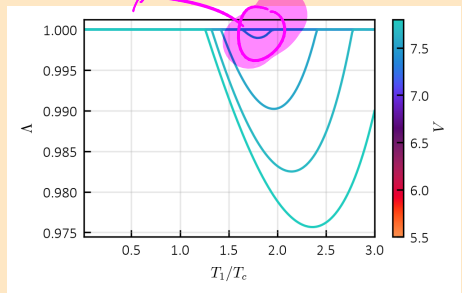
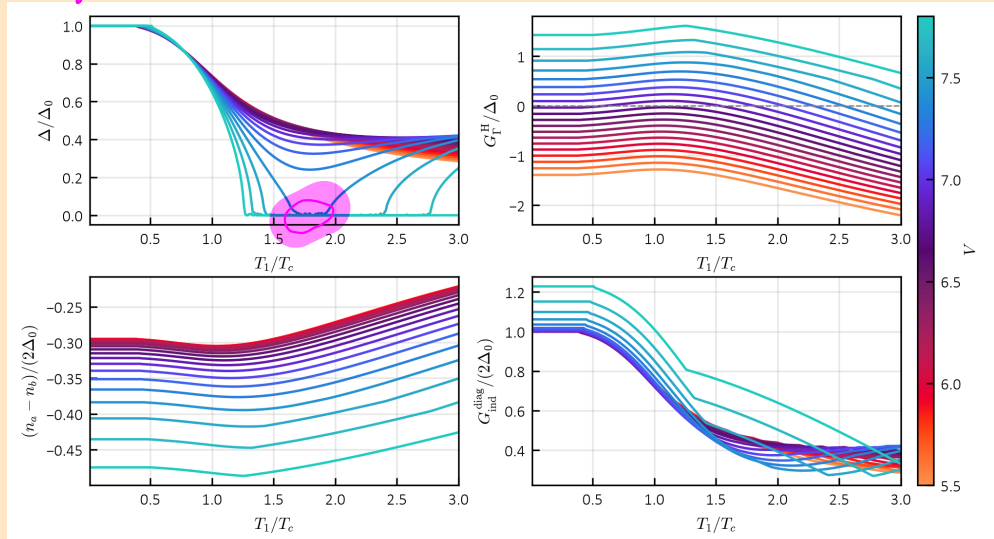
→ ko $\Lambda > 1$: normal state je unstable

→ $\Lambda = 1$ transition / kritična / fixed point

→ $\Lambda < 1$: $\Delta \rightarrow 0$, normal state.

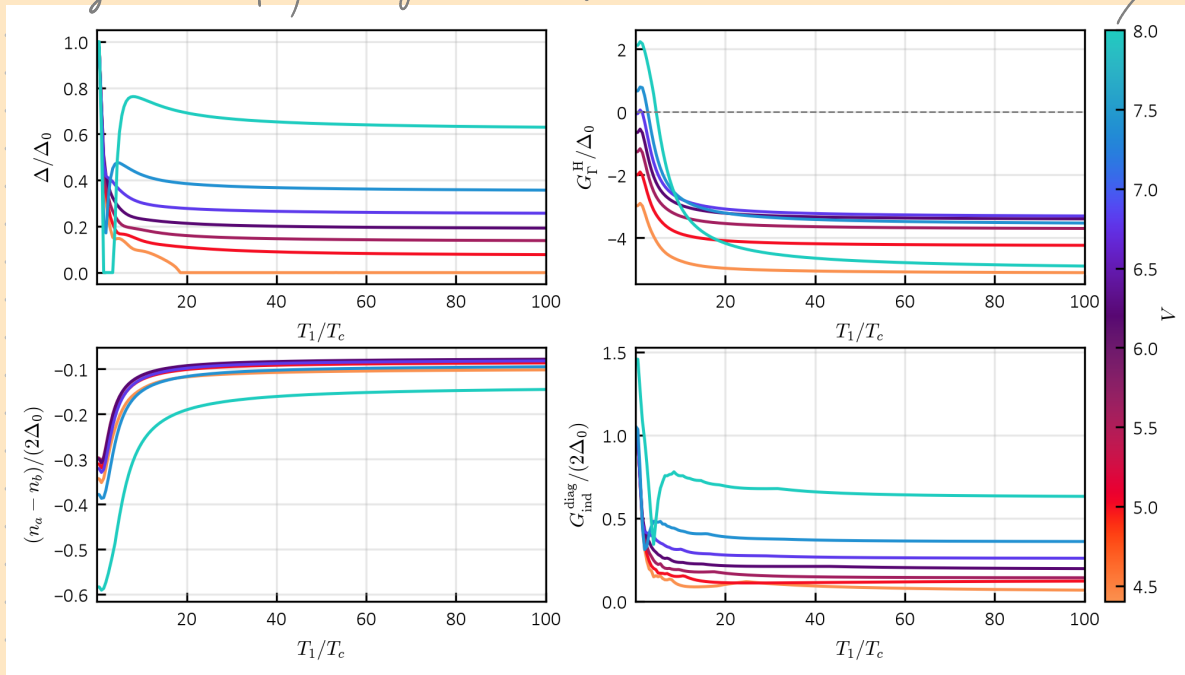
→ $|\Delta' - \Delta| = |\Lambda - 1| |\Delta|$ = gap residual, glej, da je za $|\Lambda - 1| \sim 0.1$ $|\Delta|$ nestabilen na $\sim 10^{-5}$.

$T_2 = 0.1 T_c$ fixed.

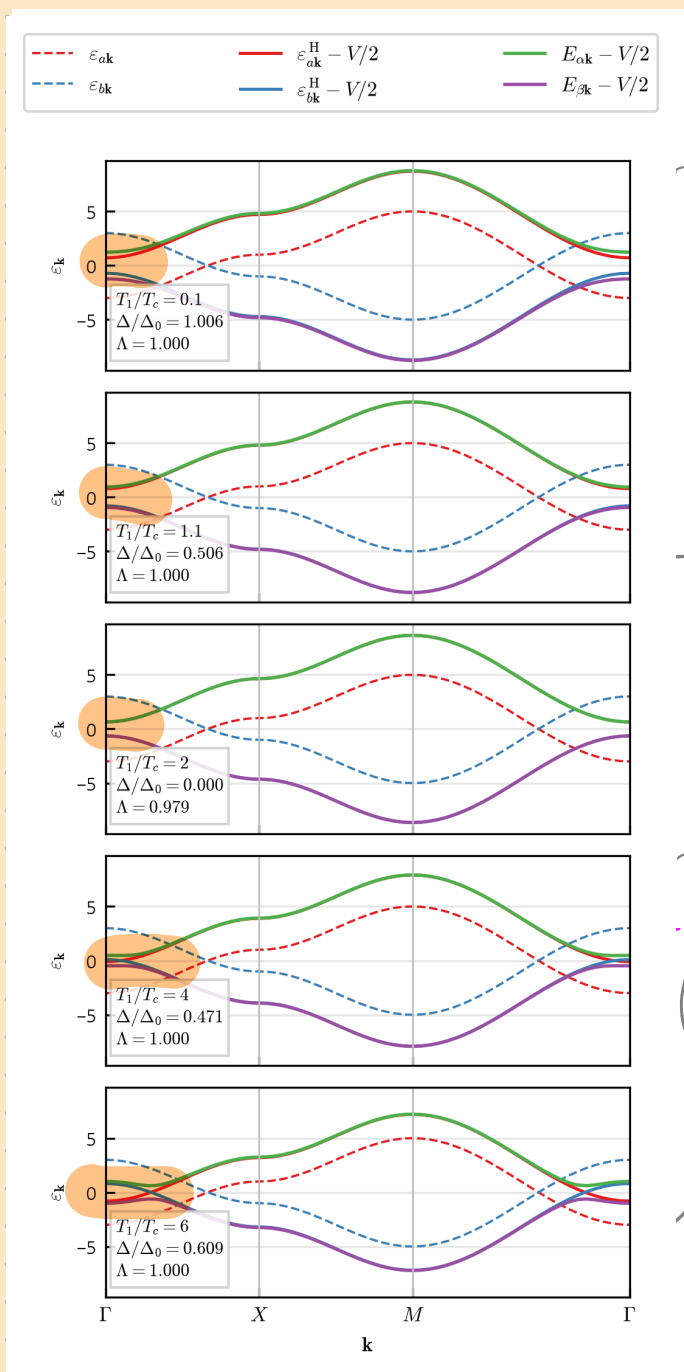
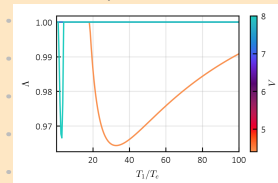


zakaj? $G^H = G^0 - V m$ ko drzujemo T_1 : $n_\beta - n_\alpha \rightarrow$ se manjša $\rightarrow \Lambda < 1 \rightarrow \Delta \rightarrow 0$
 ↓
 kerati: $m \uparrow \rightarrow G^H \downarrow \rightarrow$ manjši band separation $\rightarrow \Delta > 0?$
 $G^0 = -6$ na ryongjih grafih.

long sweep, D izgleda ravnino do $T_1 \rightarrow \infty$



v BCs režim
dijametro padne
0, ker $\Lambda < 1$



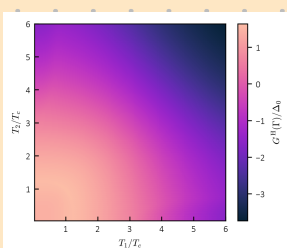
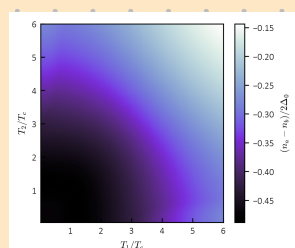
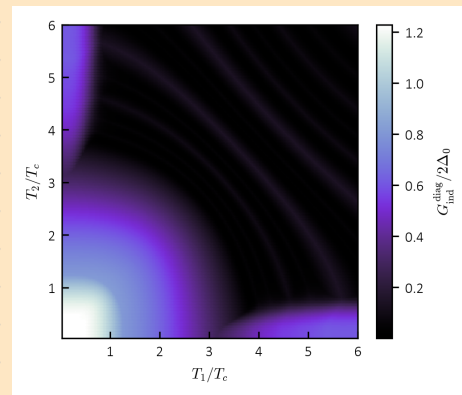
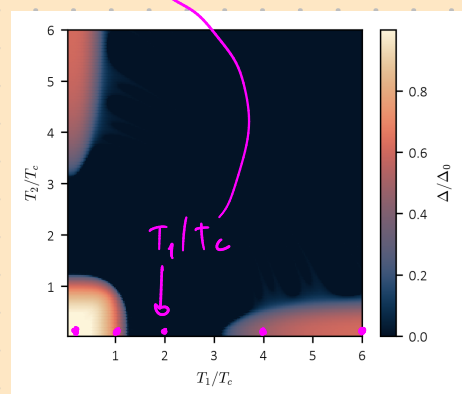
disperzija along $\Gamma-X-M$, za

$G=2$, $V=7.8$ pri $T_2=0.01T_c$
in različnih T_1 .

v pram otoku

$\Delta = 0$

reemergent
D, ker se
banda parietna
skupaj
(Narrow)



(4) equivalent T: (1) $\beta(0) \rightarrow n_{dk}(t) = \frac{1}{1 + \exp(\beta(t) E_{dk})} = f(E_{dk} \beta)$
 $(E - \frac{U}{2})?$

(2) $\langle \dot{H} \rangle = \sum_{k,j} E_{dk} \dot{n}_{dk} = \sum_{k,j} E_{dk} \sum_p [\Gamma_{kp}^{w)} n_{dp} (1 - n_{dk}) - \Gamma_{pk}^{dw} n_{dk} (1 - n_{dp})]$

(3) $\dot{\beta} = \frac{\langle \dot{H} \rangle_T}{\langle H^2 \rangle_T - \langle H \rangle_T^2}$

(4) $\beta \rightarrow \beta + \Delta t \dot{\beta}$

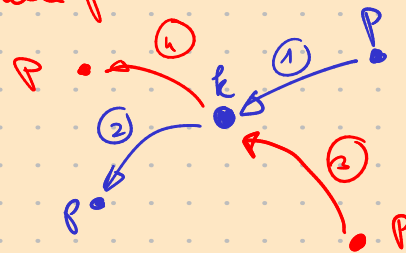
Interpretation: gley: /home/kzeleznikar/IJS-F1/Notes/theory/notes_open_systems_EI.pdf za podrobnejši.

$\downarrow \omega = \alpha$ fixed: *evoli?*

(*) $\dot{n}_{dk} = \sum_p \left[\begin{aligned} &\Gamma_{kp}^{\alpha\alpha} n_{dp} (1 - n_{dk}) \quad \textcircled{1} \\ &- \Gamma_{pk}^{\alpha\alpha} n_{dk} (1 - n_{dp}) \quad \textcircled{2} \\ &+ \Gamma_{kp}^{\alpha\beta} n_{\beta p} (1 - n_{dk}) \quad \textcircled{3} \\ &- \Gamma_{pk}^{\beta\alpha} n_{dk} (1 - n_{\beta p}) \quad \textcircled{4} \end{aligned} \right]$

delec α
 delec β

(p, k sistem)

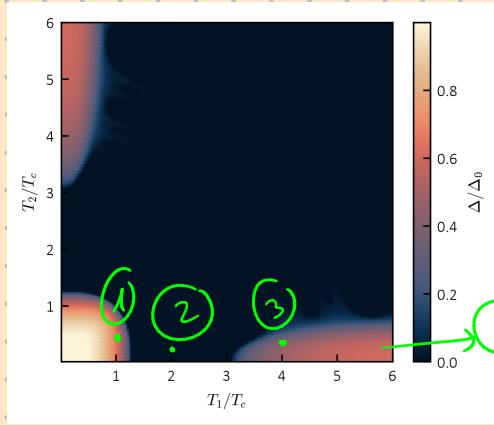


① $\propto$ iz $p \rightarrow k$ (ic ni zmedu)

② $\propto$ iz $k \rightarrow p$ (ic h j)

→ analiza fazdenesi delov za $q=2$

(A) BEC režim ($v=7.8$)



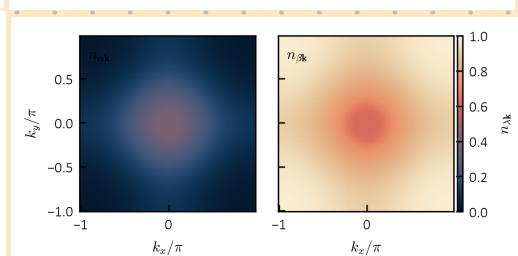
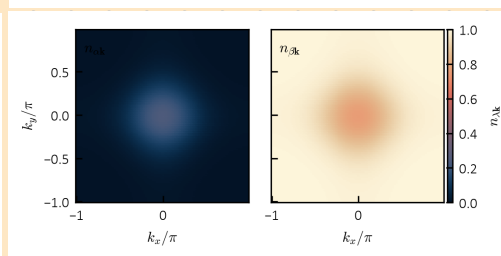
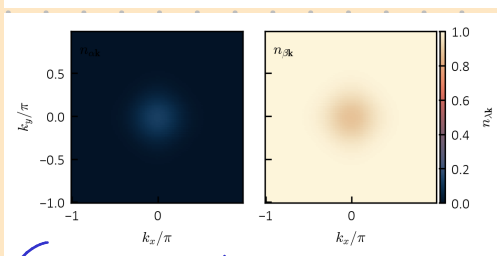
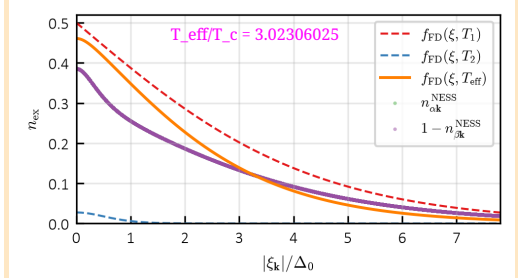
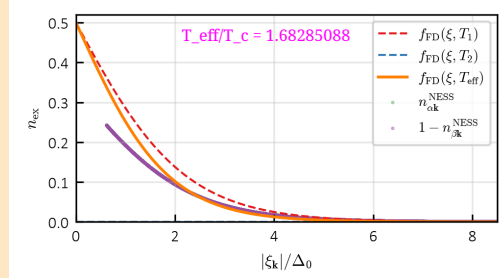
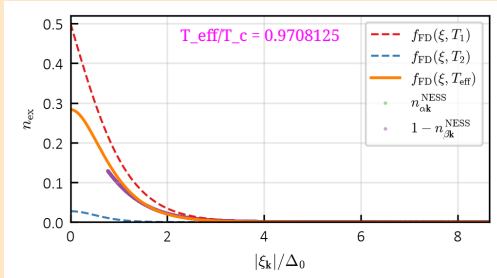
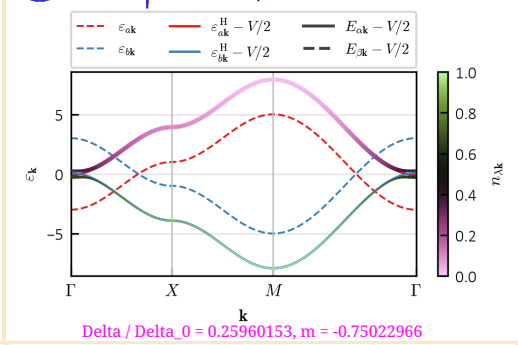
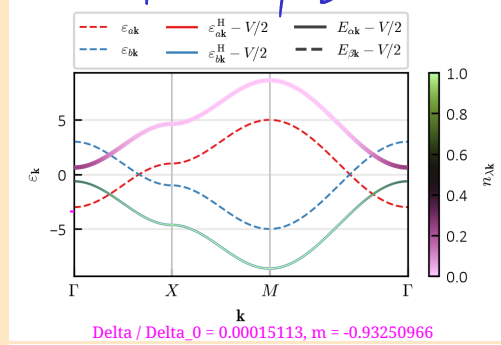
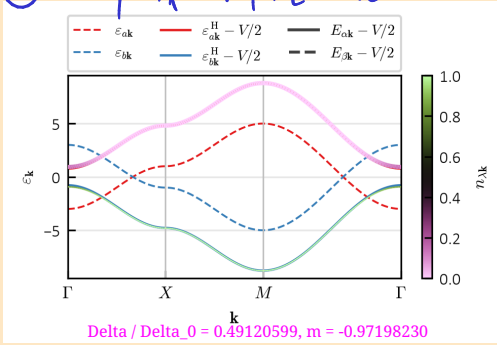
$$(\text{opomba } \xi_k = \frac{E_a(k) - E_v(k) - v(n_a - n_b)}{2})$$

$$E_{ak} - \frac{V}{2} = \frac{E_{ak} + \xi_k}{2} \pm \sqrt{\xi_k^2 + D^2}$$

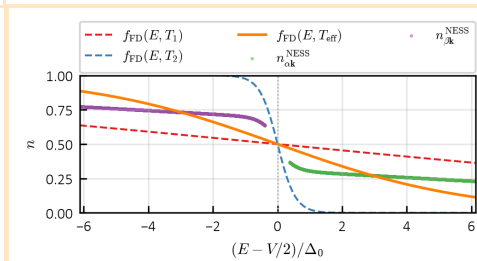
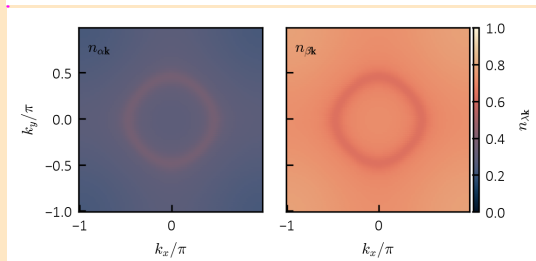
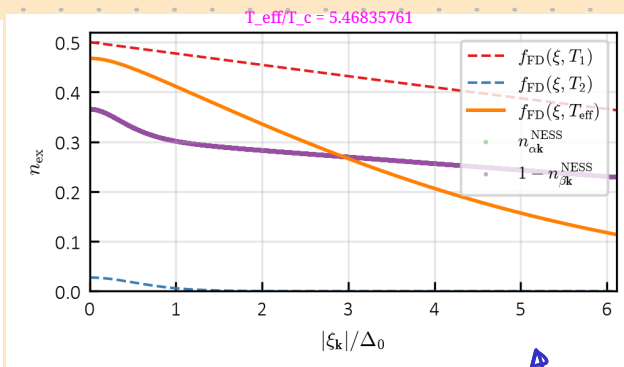
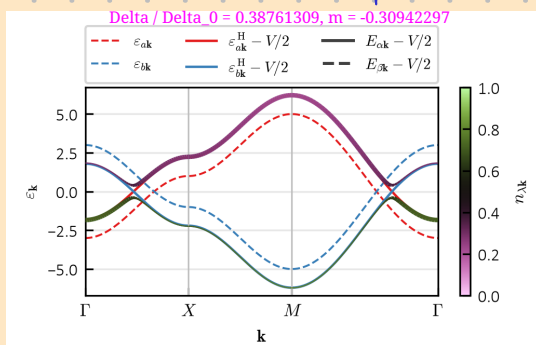
① BEC / $T_1 = 1.1 / T_2 = 0.5$

② $D=0$ / $T_1 = 2 / T_2 = 0.2$

③ BCS / $T_1 = 4 / T_2 = 0.5$



(4) BCS / $T_1 = 20 / T_2 = 0.5$

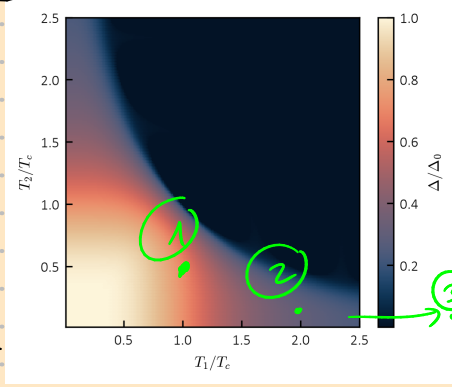


isto, drugi
variable!

→ ko $T_1 \rightarrow \infty, T_c \rightarrow 0$ gre

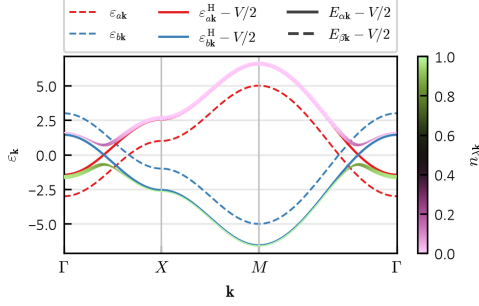
$n_k \rightarrow$ $1 < 1$

(B) BCS limit ($V=4.4$, $g=2$)



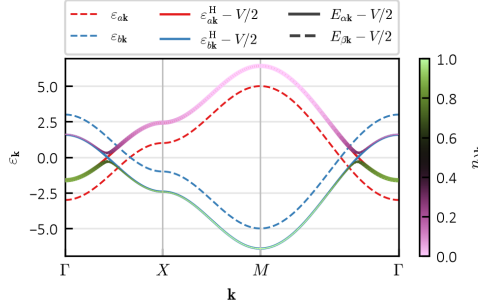
1) $T_1 = 1.1$ / $T_2 = 0.5$

Delta / Delta_0 = 0.62227223, m = -0.70813060



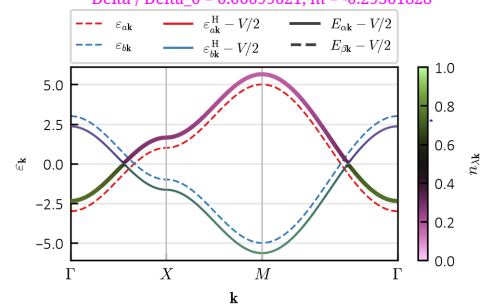
2) $T_1 = 2$ / $T_2 = 0.2$

Delta / Delta_0 = 0.29625834, m = -0.64265174

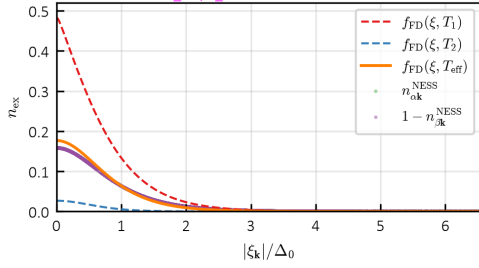


3) $T_1 = 10$, $T_2 = 0.1$

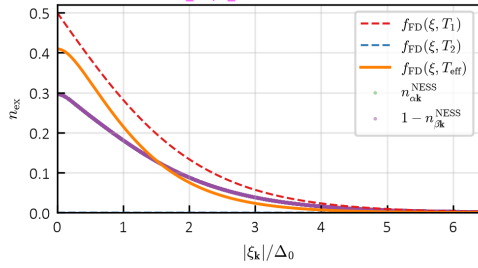
Delta / Delta_0 = 0.06699621, m = -0.29301828



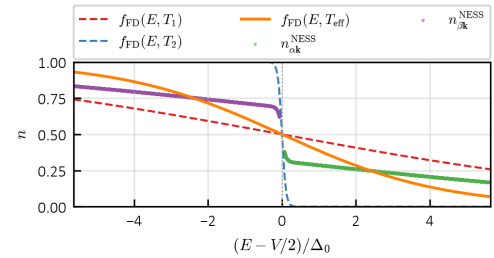
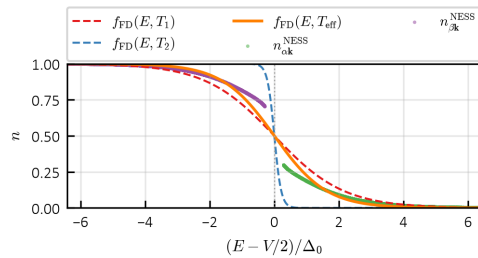
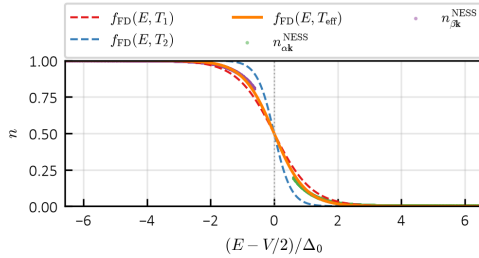
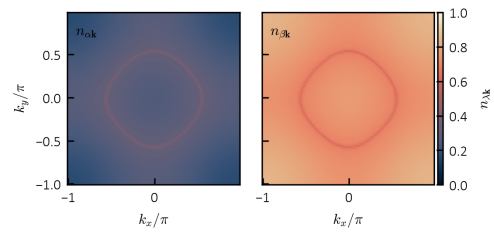
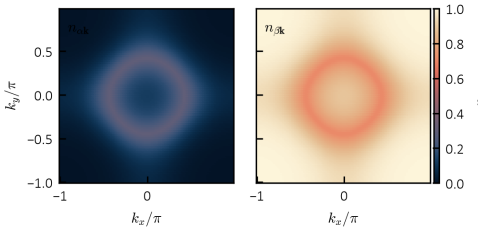
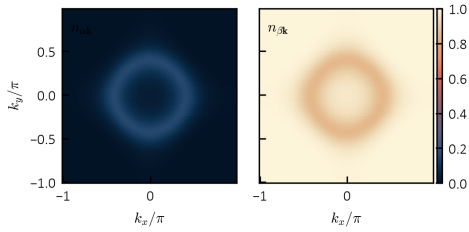
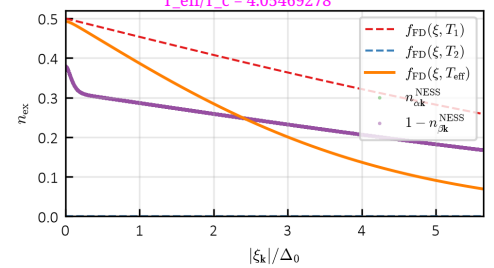
T_eff / T_c = 0.91613618



T_eff / T_c = 1.50874127



T_eff / T_c = 4.05469278



Technical notes:

Kopel za vse ploste uporablja oblike: $f_{kp}^{(\mu\nu)} = 0.1 \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}^{\alpha\beta} \cdot f\left(\frac{E_{\mu\nu} - \frac{V}{2}}{T}\right) f\left(-\frac{E_{\nu\mu} - \frac{V}{2}}{T}\right)$

in $L_{kp}^{(\mu\nu)} = f_{kp}^{(\mu\nu)} d_{\mu k}^+ d_{\nu p}$ (number conserving bath)

→ tok tokrega L je enak (*)

oznake:

$$H^{MF} = E_0 + \sum_k (c_{a,k}^+, c_{b,k}^+) \begin{pmatrix} E_{a,k} + \frac{V}{2}n - \frac{V}{2}m & -\Delta^* \\ -\Delta & E_{b,k} + \frac{V}{2}n + \frac{V}{2}m \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix}$$

$$\hookrightarrow E_{a,k} = \epsilon_k \pm \sqrt{\epsilon_k^2 + \Delta^2}; \quad \epsilon_k = \frac{E_a + E_b + V \cdot n}{2} \quad \xi_k = \frac{E_a + E_b - V \cdot m}{2}$$

$$E_{a,b}^{\pm} = E_{a,b} + \frac{V}{2}n \mp \frac{V}{2}m$$

